

MBAI 5310G-001: AI Programming

Assignment 4: Decision Tree Model
AdVantage Growth Studio: Marketing Campaign

Field	Detail
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1. Business Overview

AdVantage Growth Studio is a digital marketing agency that runs multi-channel campaigns for small and mid-sized businesses. The studio designs, executes, and measures email, social, and display campaigns on behalf of its clients. A core challenge for any marketing agency is budget efficiency: campaigns sent to the wrong customers are wasted spend, while customers who would have converted but were not targeted represent missed revenue.

This assignment builds a Decision Tree classification model to predict which customers in the studio's database are likely to convert (make a purchase) after receiving a campaign. A customer who converts generates revenue for the client; a customer who does not convert represents a wasted campaign contact. The model allows AdVantage to score and prioritize its customer list before campaign deployment.

Attribute	Detail
Business	AdVantage Growth Studio
Objective	Predict which customers will convert after receiving a marketing campaign
Target Variable	Converted (Yes / No)
Dataset Size	600 records (595 after removing 5 duplicates)
Number of Features	18 input features
Priority Metric	F1-Score, balances wasted spend (FP) and missed revenue (FN)
Model Used	Decision Tree Classifier (sklearn, random_state=42)

2. Data Preparation

2.1 Dataset Overview

The dataset contains 600 records (595 after cleaning) representing customers who received a marketing campaign. Each record has 18 features across four categories:

- Customer Demographics: Age, Annual_Income, Region, Gender
- Campaign Information: Campaign_Type, Campaign_Channel, Discount_Offered
- Engagement Behavior: Email_Opened, Email_Clicked, Time_Spent_On_Website, Social_Media_Engagement
- Purchase History: Loyalty_Score, Days_Since_Last_Purchase, Previous_Purchases, Avg_Order_Value

2.2 Preprocessing Steps

- Dropped Customer_ID, unique identifier with no predictive value
- Dropped Campaign_Date, a date string that adds no signal in this format (no time series modeling applied)
- Removed 5 duplicate records

- Confirmed no remaining missing values
- Applied one-hot encoding (pd.get_dummies) to all categorical features
- Applied Label Encoding (LabelEncoder) to the target: Yes = 1, No = 0
- Split data 80/20 using train_test_split with stratify=y and random_state=42

3. Decision Tree Model

3.1 How Decision Trees Make Predictions

A Decision Tree is a supervised machine learning algorithm that recursively partitions the data by asking binary questions about feature values. At each node, the algorithm selects the feature and threshold that best separates customers who converted from those who did not, using Gini impurity as the splitting criterion.

For AdVantage Growth Studio, the tree learns rules like: 'If Loyalty_Score > 7.2 AND Days_Since_Last_Purchase < 30, predict Converted = Yes.' These rules are directly interpretable, a campaign manager can read the tree and understand exactly what the model is responding to.

3.2 Decision Tree Visualization

The tree below shows the first three levels of the trained model. Each node displays the splitting feature and threshold, Gini impurity score, sample count, and class distribution. Orange-tinted nodes lean toward No (did not convert); blue-tinted nodes lean toward Yes (converted). Darker shading indicates a purer node, a stronger majority of one class.

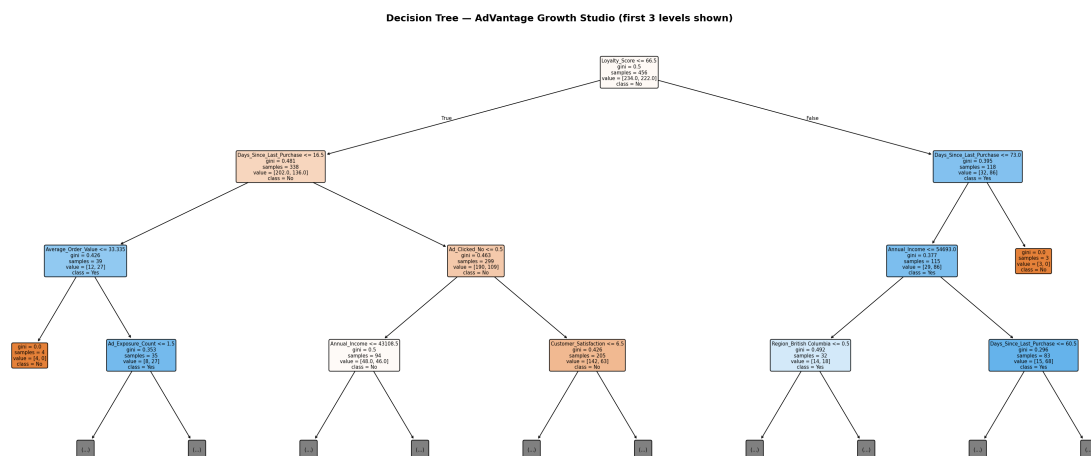


Figure 1: Decision Tree, AdVantage Growth Studio (first 3 levels shown)

4. Model Evaluation

4.1 Confusion Matrix

The confusion matrix below shows the breakdown of predictions on the test set (114 records). The rows represent the actual class, what the customer actually did. The columns represent the predicted class, what the model said they would do.

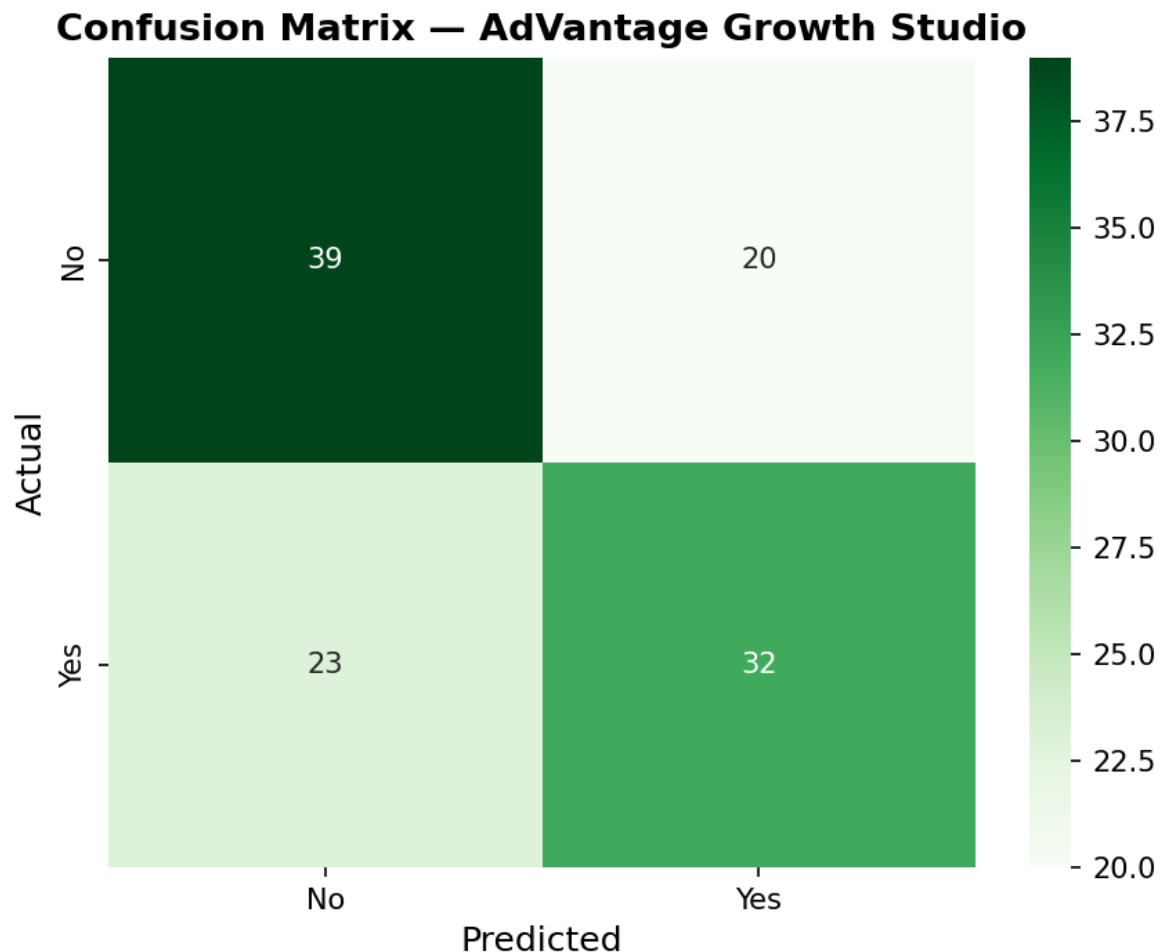


Figure 2: Confusion Matrix, AdVantage Growth Studio (unconstrained tree)

Reading the matrix:

- Top-left (39) - True Negatives: The model correctly predicted 39 customers would not convert. These customers were rightly excluded from the campaign, saving budget.
- Top-right (20) - False Positives: The model predicted 20 customers would convert, but they did not. These represent wasted campaign spend, money spent reaching customers who had no intention of buying.
- Bottom-left (23) - False Negatives: The model predicted 23 customers would not convert, but they actually would have. These are missed revenue opportunities, customers who were ready to buy but were never reached.

- Bottom-right (32) - True Positives: The model correctly identified 32 customers who converted after receiving the campaign. These are the successful, high-value campaign contacts.

Both false positives and false negatives are costly here, one wastes money, the other loses revenue. This is exactly why F1-Score is the priority metric: it penalises both types of errors equally, rather than optimising for just one.

4.2 Classification Metrics

Metric	Result
Training Accuracy	100.0%
Testing Accuracy	62.3%
Precision (Yes: converted)	62.0%
Recall (Yes: converted)	58.2%
F1-Score (Yes: converted)	60.0%
False Positives (campaigned but did not convert)	20
False Negatives (would convert but not targeted)	23

4.3 Why F1-Score is the Priority Metric

Both types of errors are costly in this business context. False positives (predicting a customer will convert but they don't) waste campaign budget, the studio spends money contacting customers who don't respond. False negatives (predicting No when the customer would have converted) represent missed revenue for the client. F1-Score equally penalizes both errors, making it the most appropriate metric here.

4.4 Overfitting Analysis

The unconstrained model achieves 100% training accuracy but only 62.3% on the test set, a gap of approximately 37.7 percentage points. This is severe overfitting. The tree memorized patterns specific to the 480 training examples rather than learning general conversion signals that apply to new customers.

5. Controlling Complexity with max_depth

5.1 Applying max_depth=4

Retraining with max_depth=4 limits the tree to four levels of splits, preventing it from fitting noise in the training data. This produces a simpler, more generalizable model.

Confusion Matrix — AdVantage Growth Studio (max_depth=4)

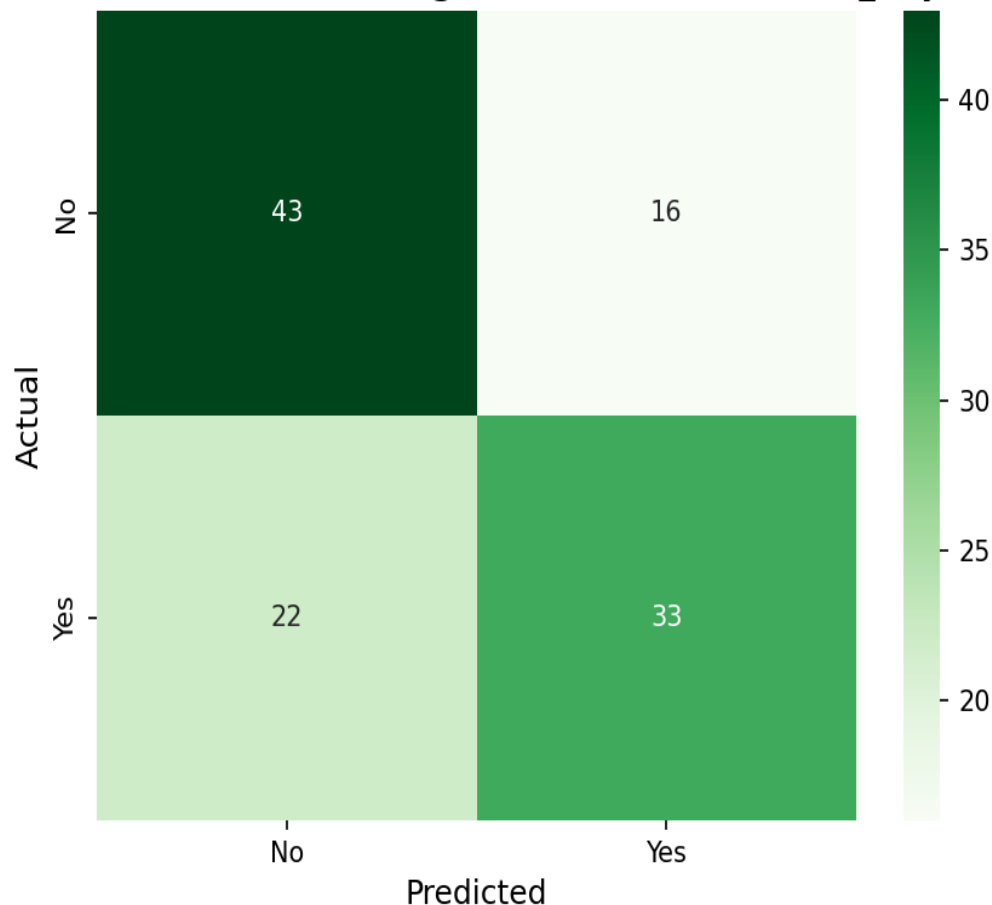


Figure 3: Confusion Matrix, AdVantage Growth Studio (max_depth=4)

Reading the max_depth=4 matrix:

- True Negatives: Customers correctly excluded from the campaign, budget was not wasted on them.
- False Positives: Customers incorrectly included in the campaign who did not convert, wasted spend, but reduced compared to the unconstrained model.
- False Negatives: Customers incorrectly excluded who would have converted, missed revenue, though the model now makes fewer of these errors.
- True Positives: Customers correctly identified as likely converters and successfully reached, the direct return on the campaign.

Constraining the depth improved test accuracy from 62.3% to 66.7%. By forcing the tree to rely on fewer, more generalizable rules, it became better at identifying real conversion signals rather than memorising quirks in the training data.

Comparison	Unconstrained	max_depth=4
Training Accuracy	100.0%	—
Testing Accuracy	62.3%	66.7%
Train/Test Gap	~37.7 pts	~33.3 pts

6. Feature Importance

The feature importance chart below shows which inputs had the greatest influence on the model's predictions. Features with higher importance scores caused larger reductions in Gini impurity across all splits in the tree.

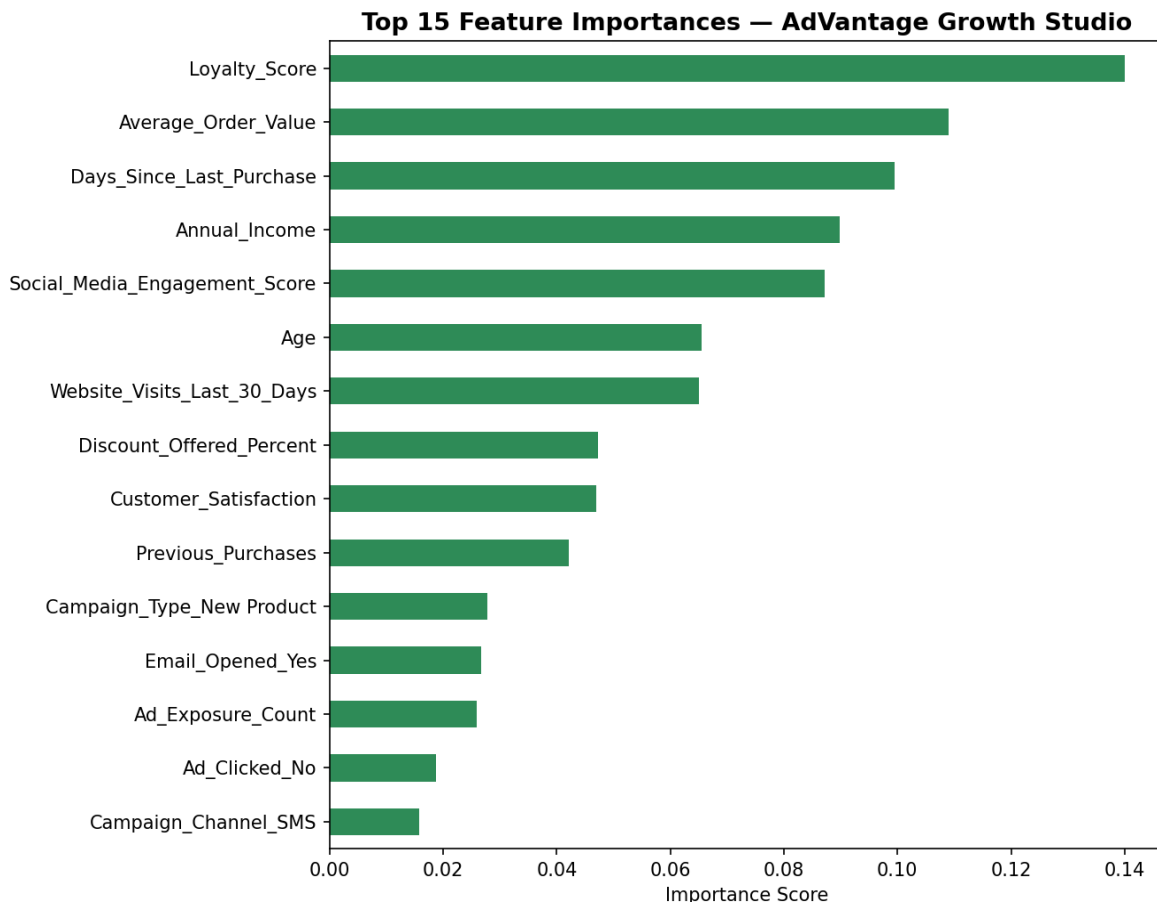


Figure 4: Top 15 Feature Importances, AdVantage Growth Studio

6.1 Business Interpretation of Top Features

Feature	Business Interpretation
Loyalty_Score	Loyal customers have an established relationship with the brand and are more likely to respond positively to campaigns. High loyalty score customers should be prioritized.
Average_Order_Value	Customers with a higher average order value are already accustomed to spending more per transaction. They are receptive to premium offers and represent higher revenue per conversion.
Days_Since_Last_Purchase	Customers who purchased recently are in an active buying mindset. Targeting customers who bought within the last 30 days significantly increases conversion likelihood.
Annual_Income	Higher income customers have more purchasing power and are more likely to convert, though this feature also raises fairness concerns if it excludes lower-income customers unfairly.
Social_Media_Engagement_Score	Customers who actively engage with social media content are more responsive to digital campaigns. High engagement scores signal openness to brand interaction and conversion.

7. Business Actions and Recommendations

Model Insight	Recommended Business Action
Loyalty_Score is the top predictor	Build a loyalty segment in the campaign list. Customers with loyalty scores above the median should receive priority placement and more personalized offers.
Recent buyers convert at higher rates	Set up a post-purchase campaign window (e.g. 7-30 days after last purchase) to reach customers while they are still in a buying mindset.
Email click behavior is highly predictive	Track email click-through rates per customer and use this as a real-time conversion signal. A/B test subject lines to increase click rates and improve model input quality.
F1-Score is 60.0%	The model has moderate predictive value but is not yet strong enough for fully automated targeting. Use its scores as a ranking tool to prioritize manual review of high-probability customers rather than automated sends.
Annual_Income raises fairness concerns	Run a fairness audit segmented by income band before deployment. Ensure lower-income customers are not systematically excluded from campaigns even when conversion probability is similar.

8. Limitations

- **Synthetic dataset:** The dataset was synthetically generated. Conversion patterns in real campaign data will be more complex and less evenly distributed.
- **Small sample size:** 595 records is insufficient for a production-grade model. A dataset of several thousand records per campaign type would be needed for reliable generalization.
- **Annual_Income and Region risk demographic bias:** The model may learn to deprioritize lower-income customers or customers from certain regions, even when their true conversion probability is similar to other groups.
- **Campaign_Date was excluded:** Time-based patterns (seasonal trends, day-of-week effects) were not captured. Incorporating temporal features could improve model performance significantly.
- **Binary prediction only:** The model returns Yes/No. A probability score (using predict_proba) would be more useful for ranking customers by conversion likelihood and setting thresholds by campaign budget.

9. Responsible AI Considerations

- Fairness audit required: Before any real-world deployment, the model must be tested for disparate impact across gender, income band, and region. Campaign targeting that systematically excludes any demographic group could violate consumer protection regulations.
- Human review before deployment: Campaign managers should review the model's top-ranked customer segments before any send. The model should support, not replace, human decision-making.
- Explainability: AdVantage must be able to explain to clients why specific customers were targeted. The Decision Tree structure supports this, the decision path for each customer prediction can be extracted and logged.
- Data retention and consent: Customer behavioral data (email opens, clicks, website time) must be collected and stored with explicit consent under applicable privacy laws (e.g., CASL in Canada, GDPR in Europe).
- Model drift: Campaign behavior changes over time. The model should be retrained at least quarterly on fresh campaign data to maintain accuracy.

10. Final Conclusion

This assignment applied a Decision Tree classifier to a marketing conversion prediction problem for AdVantage Growth Studio. The model was trained on 595 cleaned customer records with 18 input features, using an 80/20 stratified train/test split.

The unconstrained tree overfit severely, 100% training accuracy versus 62.3% test accuracy, a gap of 37.7 points. Applying `max_depth=4` improved test accuracy to 66.7% and reduced overfitting. The most important predictors were `Loyalty_Score`, `Days_Since_Last_Purchase`, and `Annual_Income`, all behavioral or economic signals that intuitively reflect a customer's readiness to convert.

F1-Score was selected as the priority metric because both false positives (wasted campaign spend) and false negatives (missed revenue) carry meaningful business cost. At 60.0% F1, the model shows moderate predictive value but is not yet strong enough for fully automated campaign targeting. It should be used as a ranking and prioritization tool, with human review of the top-scored segments before deployment.

A fairness audit is required before real-world use, particularly for the `Annual_Income` and `Region` features. Future work should explore ensemble methods (Random Forest, Gradient Boosting) and incorporate time-based features derived from `Campaign_Date` to capture seasonal and recency patterns.